SHUQIN DONG

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RESEARCH INTERESTS

Biomedical radar system design, Data Fusion, Radar and acoustic signal processing algorithm, Machine learning

WORK EXPERIENCE

Postdoctoral Fellow in City University of Hong Kong, HK SAR 11/02/2025 - present

- Radar-based smart multimodal system design.

Chief Scientist / co-Funder PetHealthAI Limited, HK / Shenzhen 11/04/2026 - present

- Multimodal monitoring for animals and personalized AI agent for healthcare.

Acoustic algorithm engineer in HUAWEI, China 5/5/2020 - 1/4/2021

- Development of a detection solution for minor audio defects based on high-precision microphones.

EDUCATION

• **D.Eng. in Shanghai Jiao Tong University** 6/4/2021 - 15/12/2024
Thesis title: Research on key technologies of bio-radar based physiological signal detection and their applications in medical clinics, advised by Prof. Changzhan Gu.

• **Visiting Ph.D. student in National University of Singapore** 31/12/2023 - 16/5/2024
Department of Electrical and Computer Engineering, advised by Prof. Yongxin Guo.

• **M.S. in Zhejiang University, China** 16/9/2017 - 30/3/2020
Thesis title: A Remote Detection of Human Heart Activity based on Doppler Radar, advised by Prof. Lixin Ran.

• **B.S. in Xidian University, China** 24/8/2013 - 13/6/2017
Thesis title: Simulation and Design of Wideband Antenna Array based on Tight Coupling Technology, advised by Prof. Wei Hu.

AWARDS

First Prize in 12th Hong Kong University Student Innovation and Entrepreneurship Competition	2026, ITC, HK
Third Prize in IMBioC 2025 Young Professionals Competition	2025, IEEE MTT-S, USA
IEEE TIM Outstanding Reviewer	2024, IEEE IM-S, USA
National Scholarship (for Ph.D. candidate, top 0.2%)	2024, 2022, Ministry of Education, China
Shanghai Outstanding Graduate	2024, Shanghai, China
IEEE MTT-S Graduate Fellowship for Medical Applications	2023, IEEE MTT-S, USA
IEEE MTT-S IWS FLASH Competition Best Paper Award	2023, IEEE MTT-S, USA
IEEE MTT-S IWS FLASH Competition Best Paper Award	2022 IEEE MTT-S, USA
CICAI Best Demo Runner-up Award	2022, CCF, China
Quality Star in 2012 Lab	2020, HUAWEI, China
National Scholarship (for master student, top 0.2%)	2019, Ministry of Education, China

PROFESSIONAL ACTIVITIES

- **Invited speaker** of the 2026 CIE International Conference on Radar (RADAR2026) – High-Precision Microwave Physiological Dynamics Signal Detection: From Far-Field to Near-Field
- **Invited speaker** of the 2025 IEEE International Workshop on Electromagnetics (iWEM) in Hong Kong – A Novel Circuit Design for Direct Current Offset Cancellation in Interferometric Radar System
- **Session Chair** during 2025 IEEE MTT-S International Microwave Workshop Series (IMWS-AMP) in Wuxi
- **TPC member** for 2024 Asilomar Conference on Signals, Systems, and Computers.

- **First-Class Award** in the Medical Artificial Intelligence Theme Competition at the 2023 First National Digital Health Innovation Application Contest
- **Speaker** for IMS2023 workshop – Biomedical Radars for Monitoring and Diagnosis of Respiratory and Cardiovascular Diseases
- **Reviewer** for IEEE Transactions on Circuits and Systems for Video
- **Reviewer** for IEEE Journal of Biomedical and Health Informatics.
- **Reviewer** for IEEE Journal of Electromagnetics, RF, and Microwaves in Medicine and Biology.
- **Reviewer** for IEEE Transactions on Microwave Theory and Techniques.
- **Reviewer** for IEEE Transactions on Instrument and Measurement.
- **Reviewer** for IEEE Microwave and Wireless Technology Letters.
- **Reviewer** for IEEE Transactions on Industrial Informatics.
- **First-Class Award** of Digital Medical Technology and Application Innovation Competition in 2022, Ruijin Hospital & Shanghai Artificial Intelligence Laboratory

PATENTS

- **Shuqin Dong** , Zhaojin Guo, Li Luv, Kai Liu, Yongxin Guo. A WEARABLE ANIMAL VITAL SIGN MONITORING SYSTEM AND METHOD FOR MONITORING A PHYSIOLOGICAL PARAMETER OF AN ANIMAL USING A WEARABLE NEAR-FIELD RADAR SYSTEM, US Provisional Patent Application No.64/051,411
- **Shuqin Dong** , Lixin Ran. A non-contact method for measuring cardiac rhythm. Chinese authorized invention patent: CN110115585A, 2019.08.13

BOOK CHAPTER

- Zhang, Y.; Dong, S. et al. Hand Gesture Recognition Based on SIMO Doppler Radar Sensors. IET, 2019. ISBN 9781785617607. Co-first-author: Open research data, software and algorithm repositories.

PROJECT EXPERIENCE

Millimeter-Wave Non-Invasive Health Visualization Monitor Principal Investigator 2025 - 2026

- Funded by HK Tech 300 Seed Fund and HONG KONG SCIENCE AND TECHNOLOGY PARKS, HK

Doppler Cardiogram Detection based on radar Principal Investigator 2023 - 2024

- Funded by MTT-S, USA
- Design and develop high-sensitivity and high-linearity millimeter-wave radar sensor system iwth linear and robust radar baseband demodulation algorithm and Doppler cardiogram extraction algorithm.

Non-contact heart rhythm and respiration detection system based on millimeter wave sensing Project participant 2023 - 2025

- Funded by Shanghai Jiao Tong University Medical and Engineering Interdisciplinary Research Fund, China

Noncontact Mornitoring and managment of chronic respiratory and cardiovascular disease using millimeter-wave Project participant 2021 - 2023

- Shanghai Jiao Tong University Artificial Intelligence Research Institute
- Selected for the 2023 World Artificial Intelligence Conference SAIL Award TOP30 list

PUBLICATIONS

Peer-reviewed scientific journals:

[1] **Shuqin Dong**, Kun Duan, Alex Wong and Yongxin Guo. *Contactless Jugular Venous Pulse Waveform Detection With a Metalens-Integrated Near-Field Bioradar* , Under review, IEEE Transactions on Instrumentation and Measurement, 2026.

[2] **Shuqin Dong**, Wenren Zhou, Xuelin Kong, Shengjie Pan, Bo Wang and Yongxin Guo. *Remote Asynchronous Phase Detection of Thoraco-abdominal Breathing: a Novel Human Backscattering Model with FMCW Radar*, IEEE Transactions on Microwave Theory and Techniques, vol. 74, no. 5, pp. 4625-4637, May 2026.

[3] **Shuqin Dong**, Li Wen, Qing Cao, Kang Chen, Changzhan Gu* and Junfa Mao. *A Reliable Framework for Contactless Velocity Cardiogram Detection With Interferometric Biomedical Radar*, SCIENCE CHINA Information Sciences, 69, 122302, January 2026.

[4] Zhi Zheng, **Shuqin Dong**, Haotian Shi, Bo Wang, Zhiying Zhou, and Yongxin Guo*. *Contactless High Dynamic-Range Blood Pressure Estimation by Modified Up-Down Stroke Time based on Doppler Wrist Pulse*, IEEE Transactions on Microwave Theory and Techniques, vol. 73, no. 8, pp. 4273-4285, Aug. 2025. **(Co-first author)**

[5] Yuchen Li, **Shuqin Dong**, C Gu. *Feel Your Heartbeat Across Space: Radar-Based Cardiac Activity Monitoring for Healthcare Applications*, Accepted by IEEE Microwave Magazine, vol. 25, no. 12, pp. 91-111, Dec. 2024. **(Co-first author)** (IEEE MTT-S Newsletter Featured Article).

[6] **Shuqin Dong**, L Wen, Y Yang, Z Zhang, Y Wang, Z Liu, Q Cao, Y Xu, C Li, C Gu. *A Review on Recent Advancements of Biomedical Radar for Clinical Applications*, IEEE Open Journal of Engineering in Medicine and Biology, vol. 5, pp. 707-724, 2024. **(Invited paper)**

[7] **Shuqin Dong**, Y Li, C Gu, J Mao. *Robust Cardiac Timing Detection Technique with Vectors Analytic Demodulation in Doppler Cardiogram Sensing*, in IEEE Transactions on Microwave Theory and Techniques, vol. 72, no. 8, pp. 4866-4877, Aug. 2024.

[8] **Shuqin Dong**, L Wen, Y Li, J Lu, Z Zhang, C Yuan, C Gu. *Remote Respiratory Variables Tracking With Biomedical Radar System During Sleep*, IEEE Internet of Things Journal, vol. 11, no. 11, pp. 19937-19948, 1 June 2024.

[9] **Shuqin Dong**, Y Li, J Lu, Z Zhang, C Gu, J Mao. *Accurate detection of Doppler cardiograms with a parameterized respiratory filter technique using a K-band radar sensor*, IEEE Transactions on Microwave Theory and Techniques, vol. 71, no. 1, pp. 71-82, Jan. 2023. **(IEEE MTT-S Newsletter Featured Article).**

[10] **Shuqin Dong**, Y Zhang, C Ma, C Zhu, Z Gu, Q Lv, B Zhang, C Li, L Ran. *Doppler cardiogram: A remote detection of human heart activities*, in IEEE Transactions on Microwave Theory and Techniques, vol. 68, no. 3, pp. 1132-1141, March 2020.

[11] **Shuqin Dong**, L Wen, Z Zhang, C Gu, J Mao. *Contactless measurement of human systolic time intervals based on Doppler cardiograms in clinical environment*, IEEE Microwave and Wireless Components Letters, vol. 32, no. 6, pp. 796-799, June 2022.(selected as **Top50 Papers** of 2022 IEEE International Microwave Symposium, converted to MWCL)

[12] Li Wen, **Shuqin Dong**, Yi Wang, Changzhan Gu*, Zheng Tang, Zhiwei Liu, Junfa Mao. *Noncontact Infant Apnea Detection for Hypoxia Prevention with a K-band Biomedical Radar*, IEEE Transactions on Biomedical Engineering, vol. 71, no. 3, pp. 1022-1032, March 2024.

[13] Z Chen, W Xu, **Shuqin Dong**, R Ma, C Gu, X Zhang. *Radar Sensing of Displacement Motions With High Robustness Against Additive Noise*, IEEE Transactions on Microwave Theory and Techniques, vol. 71, no. 8, pp. 3678-3690, Aug. 2023.

[14] W. Xu, **Shuqin Dong**, C Gu, J Mao. *A Novel Calibration-Free Motion Sensing Technique With Single-Channel Interferometric Radars*, IEEE Transactions on Microwave Theory and Techniques, vol. 71, no. 1, pp. 446-455, Jan. 2023.

[15] Y. Zhang, **Shuqin Dong**, C Zhu, M Balle, B Zhang, L Ran. *Hand gesture recognition for smart devices by classifying deterministic Doppler signals*, IEEE Transactions on Microwave Theory and Techniques, vol. 69, no. 1, pp. 365-377, Jan. 2021.

[16] Y. Zhang, C Zhu, **Shuqin Dong**, Z. Gu, M Balle, B Zhang, C. Li, L Ran. *3-D Motion Imaging in a Multipath Coordinate Space Based on a TDM-MIMO Radar Sensor*, IEEE Transactions on Microwave Theory and Techniques, vol. 68, no. 11, pp. 4642-4651, Nov. 2020.

Peer-reviewed conference proceedings:

[1] **Shuqin Dong**, B. Wang, C. Gu and Y. Guo. *Accurate Noncontact Sleep Apnea-hypopnea Detection Using a K-band Biomedical Radar*, in 2024 15th Global Symposium on Millimeter-Waves & Terahertz (GSMM), Hong Kong, 2024, pp. 148-150.

[2] **Shuqin Dong**, J Lu, Y Chen, C Gu, J Mao, et al. *Accurate Fast Heartrate Detection based on Fourier Bessel Series Expansion Technique During Radar-based Sleep Monitor*, in IEEE/MTT-S International Microwave Symposium **(IMS)**, 2023.

[3] **Shuqin Dong**, Z Zhang, Y Li, J Lu, H Wu, C Gu, J Mao. *COVID-SENSE: Radar-Based Remote Respiratory Disorder Detection in Clinical Environment*, in IEEE MTT-S International Wireless Symposium **(IWS)**. 2023. **(FLASH Competition Best Paper Award)**

[4] **Shuqin Dong**, L Wen, C Gu, J Mao. *Contactless Detection for Heart Sounds Based on Doppler Radar Sensor*, in IEEE MTT-S International Wireless Symposium **(IWS)**. 2022. **(FLASH Competition Best Paper Award)**

[5] **Shuqin Dong**, C Gu, J Mao. *Contactless Cardiac RR Intervals Estimation in Radar-Based Cardiogram Detection*, in 56th Asilomar Conference on Signals, Systems, and Computers **(Asilomar)**. 2022. **(Finalist of Best Student Paper Competition)**

[6] **Shuqin Dong**, C Gu, X Yang. *Contactless Cardiogram Reconstruction Based on the Wavelet Transform via Continuous-Wave Radar*, in Second CAAI International Conference **(CICAI)**. 2022. **(Best Demo Runner-up Award)**

[7] **Shuqin Dong**, C Gu, Z Zhang, J Mao. *Doppler Cardiogram Detection in Clinical Environment*, in IEEE Radio and Wireless Symposium **(RWS)**, 2022.

[8] **Shuqin Dong**, C Gu, Z Zhang, J Mao. *Doppler Cardiogram Detection in Clinical Environment*, in 2022 IEEE MTT-S International Microwave Biomedical Conference **(IMBioC)**, 2022.

[9] **Shuqin Dong**, Y Zhang, C Ma, Q Lv, C Li, L Ran. *Cardiogram detection with a millimeter-wave radar sensor*, in IEEE Radio and Wireless Symposium **(RWS)**. 2020. **(Finalist of Best Student Paper Competition)**

[10] **Shuqin Dong**, Y Zhang, C Ma, Q Lv, C Li, L Ran. *A millimeter-wave Doppler sensor for bio-signals detection*, in 2018 IEEE MTT-S International Wireless Symposium **(IWS)**, 2018. **(Student Paper Competition Honorable Mention Award)**

[11] W. Li, **Shuqin Dong**, Y. Wang, C Gu, Z. Tang, Z. Liu, J Mao. *Noncontact Monitoring of Infant Apnea for Hypoxia Prevention Using a K-band Biomedical Radar*, in IEEE/MTT-S International Microwave Symposium **(IMS)**. 2023.

[12] J Liu, L Wen, **Shuqin Dong**, Z Zhang, C Gu, J Mao. *PS-BPC01-4: NONINVASIVE CONTINUOUS BLOOD PRESSURE MONITORING BASED ON WEARABLE RADAR SENSOR WITH PRELIMINARY CLINICAL VALIDATION*, in Journal of Hypertension, 41 (Suppl 1), e291-e292.

[13] W. Li, **Shuqin Dong**, Z Zhang, C Gu, J Mao. *Noninvasive continuous blood pressure monitoring based on wearable radar sensor with preliminary clinical validation*, in IEEE/MTT-S International Microwave Symposium **(IMS)**. 2022.

[14] W. Li, **Shuqin Dong**, C Gu, J Mao. *Antenna-in-Package Millimeter-Wave Sensor for Monitoring Heart Rate Variability*, in IEEE 10th Asia-Pacific Conference on Antennas and Propagation **(APCAP)**. 2022.

[15] Y. Li, J. Lu, **Shuqin Dong**, C Gu, J Mao. *Temporal-Spatial Equivalent Virtual Array Technique for Accurate Vital Sign Monitoring*, 2023 IEEE/MTT-S International Microwave Symposium **(IMS)**. 2023.

[16] L Wen, Z Zhang, J Wang, J Liu, **Shuqin Dong**, C Gu, J Mao.*Enhancing Heart Failure Monitoring: Biomedical Radar-Based Detection of Cheyne-Stokes Respiration*, 2024 IEEE Radio and Wireless Symposium **(RWS)**, 2024.

[17] W Li, Y Li, J Zhang, J Lu, **Shuqin Dong**, C Gu, J Mao. *A Feature-based Filtering Algorithm with 60GHz MIMO FMCW Radar for Indoor Detection and Trajectory Tracking*, 2023 IEEE/MTT-S International Microwave Symposium **(IMS)**. 2023.

[18] Z Chen, Y Zhang, **Shuqin Dong**, Y Zhang, J Wang, C Li, L Ran. *Wideband architecture for passive intermodulation localization*, in 2018 IEEE MTT-S International Wireless Symposium **(IWS)**, 2018.